

Measure	Description	Interpretation
Speed	The relationship between steps and the sum of the first step's ground contact time and the flight time leading to the subsequent ground contact. Alternatively, instantaneous speed or velocity can be examined using GPS technology, LiDAR, laser, radar, motorized resistance devices, and 2D and 3D motion capture.	- Speed dictates how fast an athlete can move the body mass over a specific distance. - Instantaneous speed/velocity at time points/distances can be calculated, facilitating a greater insight into sprint strategy
Acceleration	The change in speed as determined by the delta of two steps (e.g., right heel to left heel) and the sum of ground contact times and flight times of each step.	Acceleration provides insight into how well the athlete changes speed (increasing velocity in a direction) over a prescribed distance. For most athletes, furthering the length an athlete can accelerate will lead to higher top speed, provided that postural integrity and mechanics are sound.
Technique	Acceleration and maximum velocity techniques can be objectively quantified (i.e., joint angles) or qualitatively examined and compared to technical models to help improve performance and mitigate injury risk.	Qualitative analysis is relatively simple to perform and only requires filming of sprint technique. High-speed video recording is a common feature of most smartphones and tablets, and athletes' technique can be compared to technical models such as Altis Kinogram or Sprint-Mechanics assessment score (16) to help individualize training.

Note: ETG = electronic timing gate; 2D = two-dimensional; 3D = three-dimensional; LiDAR = light detection and ranging.

Agility Development Strategies

The development of agility is best achieved using a periodized programming method. Random programming or the development of agility using only sport-specific methods such as small-sided games has not been found to be as effective (13). Although agility is a constant element on the field of play or during skills practices, it is recommended that agility development begin with the use of COD drills (preplanned) and progress in difficulty through increases in the physical demands; this is followed by the addition of drills involving perceptual-cognitive stress or what are typically termed *agility drills* (60, 110). In line with this recommendation, the development of agility should start with a needs analysis based on assessment of the athlete's COD ability and agility. The agility test should specifically evaluate perceptual-cognitive ability, as previously described in [table 20.3](#) on [page 655](#). Further, depending on the additional movement requirements of the sport, one may also choose to use a test that is more dependent on maneuverability.

The following is an example of steps for determining the appropriate tests, analyzing the results, and using them to program agility development of an athlete. Because agility performance is underpinned by multiple factors, one should target the largest window of opportunity through direct assessment of the strengths and weaknesses of the athlete.

Step 1. Perform a needs analysis of the sport and match tests appropriately to assess these qualities ([table 20.9](#)).

Step 2. Determine strengths and weaknesses by comparing results as a standardized score to performance standards or team mean ([figure 20.13](#)).

Step 3. Plan the development of a primary area of need and a secondary area of need for the athlete.

Step 4. Distribute the time available for this development based on need identification.

Step 5. Provide a preliminary plan for transition of percent distribution through the training blocks ([table 20.10](#)). Determine if this progression is appropriate by monitoring the development of COD and agility as described in [table 20.8](#) (e.g., COD deficit).

TABLE 20.9 Needs Analysis for Agility Development

Sport	Basketball
Timeframe	12 weeks (8 weeks off-season and 4 weeks preseason training)
Sport-specific agility needs analysis	High-velocity COD • Backdoor cuts, cuts through the lane, fast break, and possession changes • Maneuverability (multiple modes of COD movement) • Defensive shuffling, backpedaling, and bending changes of direction around screens to lose defender • Agility (perceptual-cognitive) • Decision-making time (provides great advantages both offensively and defensively) • Ability to move in reaction to play or opponent
Recommended tests	5-0-5 including COD, T-test, agility test (decision time measured)

Note: COD = change of direction.

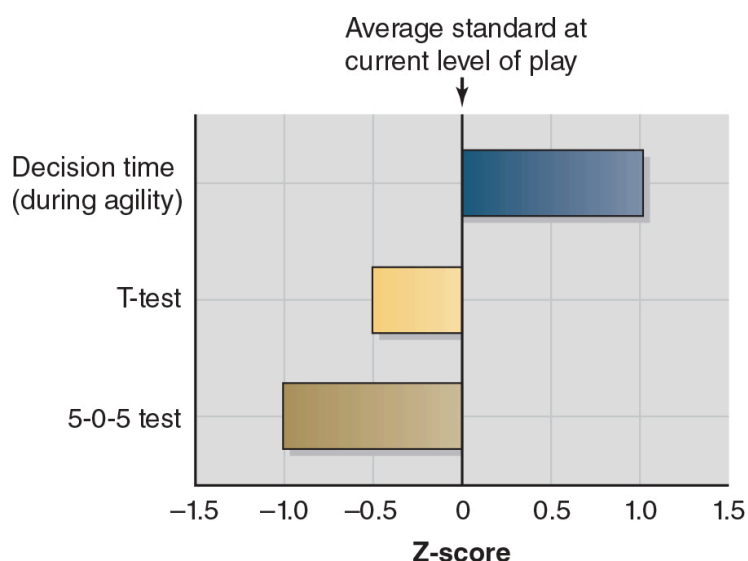


FIGURE 20.13 Strengths and weaknesses of an athlete through representation of standardized scores for perceptual-cognitive ability, maneuverability, and COD ability. Negative values indicate performance lower than average, whereas positive values indicate performance better than average.

With the information provided by the testing as well as considering other physical fitness capacities, one can implement an appropriate periodized plan. [Table 20.11](#) shows an example of three blocks of training (four weeks each). The length of time for a training session is between 10 and 15 minutes inclusive of work and rest time and can be completed one to three times per week depending on additional requirements of the athletes (e.g., skills training increasing in the preseason). A variety of drills can replace the examples given here; however, they should be chosen based on their classification and ability to address the athlete's needs.

TABLE 20.10 Planned Distribution of Training Blocks for Agility Development

	COD	Maneuverability	Agility (perceptual-cognitive)
	PRIMARY AIM	SECONDARY AIM	TERTIARY AIM
Estimated % time allotted	Block 1 ~65% Block 2 ~60%	Block 1 ~35% Block 2 ~30%	Block 1 N/A Block 2 ~20%

	COD	Maneuverability	Agility (perceptual-cognitive)
through blocks	Block 3 ~50%	Block 3 ~25%	Block 3 ~25%
Additional comments	Check strength or specifically load absorption ability with drop landings.	Assess body positioning during lateral shuffling and COD.	May be reliant on perceptual-cognitive ability. This may have previously masked the identification that the athlete could improve his or her physical COD ability.

TABLE 20.11 Application of Agility Development Strategies

Block and emphasis	Training tools and notes	Load prescription for week 1
Block 1 Primary aim: COD (deceleration capacity—linear) Secondary aim: Maneuverability (body position)	- Braking tasks are eccentric focused; therefore, start with low volumes similar to those used to introduce eccentric tasks (plyometrics). - Focus on the athlete staying low; sacrifice speed for positioning, and emphasize optimal deceleration mechanics.	2-3 sessions per week; rest period between sets COD Drill - Decelerations* (5 yd [5 m] start-stop) 1 × 6 (30 s) (Alternate nonpreferred, preferred leg finish braking [split stance] position and bilateral athletic ready position.) Maneuverability Drill - Lateral shuffling @ height or backpedaling 1 × 4 × 11 yd [10 m] (30 s)
Block 2 Primary aim: COD (deceleration capacity—lateral) Secondary aim: Maneuverability	Ensure that the athlete can use body position to effectively brake when oriented both forward and laterally; focus on the athlete absorbing into stop.	2-3 sessions per week; rest periods between sets COD Drill Decelerations* (11 yd [10 m] start-stop) 1 × 6 (30 s) as with week 1; use same variations

Note: COD = change of direction.

*Speed should progressively increase for deceleration drills through the weeks—for example, half speed to three-quarters speed entry velocity or changing distance of start-stop drills inclusive of generic and sport-specific stimuli.

Note: There is not a separate emphasis on acceleration ability here; it is expected this will be addressed in conjunction with speed sessions.

Block and emphasis	Training tools and notes	Load prescription for week 1
(transition between modes of COD—shuffle to sprint to shuffle)	Expand to a single effort in reaction to generic stimulus. Promote effective application of force into the ground (“pushing the ground away”); watch shank angle and trunk position for visual feedback.	- Decelerations (lateral) 1 × 4 (30 s) Maneuverability Drill - Z-drill (forward) 2 × 4 (30 s; 2 min) (Always alternate start direction: left to right vs. right to left direction) - Y-drill using different cutting maneuvers; side-step, crossover, or split-step to develop movement solutions. Ensure exit out of directional change is completed into a rapid deceleration. 3 × 4 (1 min; 3 min) (Generic stimuli could be added later within the block)
Block 3 Primary aim: COD (effective deceleration to reacceleration) Secondary aim: Maneuverability (combine effective transition and body positioning) Tertiary aim: Agility (focus on keeping COD performance and body position even in response to a generic stimulus)	- Focus on “braking in” and “exploding out” of the COD. - Focus on footwork and keeping low, with steps that promote a “drop and drive” out of the COD. - This is used to check the transfer of physical training to a more specific (yet still generic) stimulus. Focus on the “decide and go,” checking for athletes to reorient effectively or to move as effectively as they decide.	1-2 sessions per week (depending on skills volume); rest period between sets COD Drill - Modified 5-0-5 drill 2 × 4 (30 s; 2 min) (Alternate plant foot) Maneuverability Drill - T-test 1 × 4 (30 s) Agility - Reactive agility (Y agility drill) 1 × 4 (30 s) (Same direction in response to pointing in a direction) - Skills training will also inherently contain sport-specific stimulus training

Note: COD = change of direction.

*Speed should progressively increase for deceleration drills through the weeks—for example, half speed to three-quarters speed entry velocity or changing distance of start-stop drills inclusive of generic and sport-specific stimuli.

Note: There is not a separate emphasis on acceleration ability here; it is expected this will be addressed in conjunction with speed sessions.

Monitoring Agility and Change-of-Direction Ability

In line with the concepts considered in monitoring sprint performance, simply assessing performance modification in either COD or agility according to time to completion of the chosen test does not isolate the physical quality or perceptual-cognitive quality that one is trying to examine—the actual ability to change direction (70, 71, 73, 92, 96). Therefore, when monitoring COD or agility performance, the strength and conditioning professional may consider the measures in [table 20.12](#). Although these aspects of performance have been measured using expensive devices such as three-dimensional motion analysis equipment, readily available high-speed cameras (collecting more than 100 frames per second) could easily be used in conjunction with the usual timing methods (hand or electronic).

TABLE 20.12 Monitoring Agility Development

Measure	Description	Interpretation
Completion time	Total time to complete the task	Represents the duration it takes the athlete to get from A to B but can be biased toward athletes with superior linear speed; provides limited information regarding COD ability, technique, or strategy. Can be simply assessed using a stopwatch or ETGs.
COD deficit	The difference in time between a straight-line sprint and a COD test of equal length	Assessing the COD deficit provides insight into the additional amount of time required by the athlete due to the demands of COD. An example is subtraction of 10 m sprint time from 5-0-5 COD time (a common COD test is 10 m in length). However, COD deficit can be biased toward slower athletes and

Note: COD = change of direction; COM = center of mass; ETG = electronic timing gate; 2D = two-dimensional; 3D = three-dimensional; LiDAR = light detection and ranging.

Measure	Description	Interpretation
		provides limited information regarding COD ability, technique, or strategy.
Ground contact time (GCT)	The total time the foot is in contact with the ground during the COD (length of the plant phase, main execution plant foot contact)	Shorter ground contact times are generally associated with faster COD performance and indicate the athlete is spending less time performing the COD (if the athlete deflects their COM at the same angle). Insight into GCT can establish whether the athlete is improving their COD ability irrespective of total time. GCT can be measured using a built-in force platform, contact mat, or verified using 2D analysis.
Exit velocity	- The horizontal velocity of the athlete during the initial step out of the plant of the COD - Considered the first step of reacceleration from plant foot to foot strike	Unlike the situation for total time, using a high-speed video camera or 3D motion analysis to assess the velocity of the athlete (at the hip or COM) can give a direct measure of the ability to change direction. Improved exit velocity with the same or faster ground contact times implies greater RFD relevant to improving COD ability.
Entry velocity	Velocity of the athlete before the plant phase; can be measured over several steps or as the last step entering the plant phase of the COD	Entry velocity is a strong determinant of COD completion time, and the ability to enter the COD with high velocity can be advantageous for COD actions. Insight into entry velocity can be used to establish whether any pacing or self-regulatory strategies are adopted. Assessed using the above methods for exit velocity.
Decision-making time	Measurement of the time between two events is considered decision-making time and can either be a positive or a negative value. Decision-making time can be broken down into two different orientations to	Similar to the measurement of entry and exit velocity, a high-speed camera can be positioned to the side of the directional change to capture both the stimulus and the athlete's first definitive foot plant to change direction in response. If the decision-making time is negative, this indicates that the athlete anticipated the

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Measure	Description	Interpretation
	comprehensively assess how athletes respond during game scenarios: (a) defensive (move in the same direction as the stimulus); (b) offensive (move in the opposite direction to the stimulus).	directional change before the stimulus finalized the movement. Negative decision-making time is observed only during response to movements of another person (video or human stimulus), highlighting the importance of stimulus specificity during agility protocols. The measurement of decision-making time in both defensive and offensive situations allows strength and conditioning professionals to identify whether athletes need to develop perceptual-cognitive ability in a specific game scenario.
COD phase-specific information	Kinetic or kinematic data pertaining to the approach or acceleration, deceleration, turn, and reacceleration phases of a COD task	This enables the practitioner to examine the overall COD by splitting the task into several phases (e.g., approach, COD, and exit). This information enables the identification of COD strategy and establishes strengths and areas for development. Essentially, it provides further information regarding how the athlete achieved that completion time. Advanced technology such as LiDAR or radar may provide information regarding time and distance braking and speed at braking onset.
Alternative considerations	Practitioners should also consider the athlete's momentum into the COD task because greater momentums require greater braking impulse to decelerate. Additional monitoring of pacing strategies is also essential because athletes may regulate entry speed to reduce momentum to make deceleration or braking easier. Finally, video analysis is also	

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Measure	Description	Interpretation
	recommended to observe technique to understand compensatory strategies, areas for development, and comparison to technical models.	

Note: COD = change of direction; COM = center of mass; ETG = electronic timing gate; 2D = two-dimensional; 3D= three-dimensional; LiDAR = light detection and ranging.

Conclusion

Speed, change of direction, and agility are well-established measures of athletic prowess in most sporting contexts. For this reason, strength and conditioning professionals should be aware of how these abilities are established and modified through training plan design and execution. These components of athletic movement and game play are underpinned by task-specific force application. This task-specific force production must be large enough to stabilize the athlete's body weight in the presence of gravity, displace the athlete from one point to another, and continue moving the athlete at high velocities.

Running speed is associated with the relationship between step length and rate. Sprinting is a near-maximum to maximum run that is defined by large rates of force developed within a short stance time. Better sprinters spend more time in the air as a result of increased step lengths that occur at faster rates than those of their slower counterparts and are better able to produce larger net forces that are oriented optimally for effective acceleration and top speed. Agility encompasses the skills and abilities needed to explosively change movement direction, velocities, or modes in response to a stimulus. Both sprinting and agility use the SSC and greater neuromuscular efficiency compared to slower movements. However, agility includes perceptual-cognitive demands such as visual search scanning, decision making, anticipation, and reaction times that make it a separate training quality.

Strength and conditioning professionals aiming to improve these performance attributes should do so through the cyclic and sequential arrangement of training phases that harmonize strength training with speed, agility, or COD practice or some combination of these. This training effect may best result from the incorporation of exercises that are task specific and promote proper movement mechanics. The training prescribed should reflect the results of a needs analysis assessment at the beginning of a long phase or year.

Furthermore, subsequent training plans should be optimized through an ongoing athlete monitoring program, which can assist the strength and conditioning professional in determining how the athlete is responding to the programmed agenda.

SPEED AND AGILITY DRILLS

SPEED DRILLS

- 1 Ankling [675](#)
- 2 A-Skip [676](#)
- 3 B-Skip [678](#)
- 4 Switches [680](#)
- 5 Dribbles [681](#)
- 6 Straight-Leg Bound [682](#)
- 7 Sprint Resistance: Incline for Acceleration [684](#)

AGILITY DRILLS

- 8 Deceleration Drill [685](#)
- 9 Y Agility Drill [686](#)
- 10 Multiplanar Acceleration Initiation Box Drill [689](#)
- 11 Evasion Drill [690](#)

Speed Drills

1 ANKLING

Ankling is a commonly prescribed sprint drill to develop the correct foot contact strategy during maximum-velocity sprinting.

Starting Position

- Begin the exercise in a tall stance with the torso directly above the hips, knees, and ankles (stacked joints).

Movement Phase

- Initiate movement by pushing off one limb through the ankle, achieving a small flight phase. For the subsequent foot ground contact, the ankle should be in a neutral position and make contact on the forefoot under the hips or center of mass.
- During ground contact there should be minimal knee and ankle flexion. The athlete should be cued to “punch the ground” with the limb, emphasizing short, stiff ground contacts with each ground contact.
- The trunk should remain upright with a slight lean and hips extended (little to no flexion during ground contact). The contralateral arm range of movement will be small, but the speed of the arm movement should match the rate of foot contacts.

Technique Tips

- Ensure an upright trunk and avoid leaning back.
- Make sure the athlete drives vertically with each foot contact, emphasizing a stiff strategy landing on the forefoot with the ankle in a neutral position.
- Ensure the arm movement is timed with the leg movement and will be of a low amplitude.
- Avoid excessive hip flexion, resulting in excessive foot contact ahead of the COM, with the aim for foot contact as close as possible to under the COM.



2 A-SKIP

The A-skip is a commonly prescribed sprint drill to develop the correct leg recovery mechanics for maximum-velocity sprint running.

Starting Position

- Begin the exercise in a tall stance with the torso directly above the hips, knees, and ankles (stacked joints).

Movement Phase

- Initiate movement by lifting one leg to a flexed knee position in which the top of the thigh is near parallel. The foot of the lifted leg (swing leg) should be approximately at knee height of the stance leg, with the heel pulled up under the buttocks and the ankle dorsiflexed. This will result in the appearance of a figure four at the legs. The swing leg will always accord with this description.
- Begin the first skip by driving the swing leg down to the ground through an active foot, established through slight dorsiflexion. Lift the big toe up to establish appropriate dorsiflexion.

- Complete the drive down of the swing leg through the forefoot until near-triple extension of the new stance leg occurs.
- The fore- to mid-foot of the new stance leg should land under the hips, maintaining the appearance of stacked joints.
- At the initiation of contact, the opposing leg should pop up rapidly to create a swing leg.
- The force generated from the active push down should coincide with a skipping motion that will result in a horizontal displacement of the body down the track.

Technique Tips

- Maintain a tall posture with relaxed shoulders through the entire duration.
- The arms should move at the same rate as the legs, with minimal to no pausing of arm action between cycles.

Regression and Progression

- A regression of the drill includes an A-march, which involves the same movements but at a slower pace to establish the correct leg recovery mechanics. If athletes struggle coordinating the contralateral arm movement, the hands can be placed on the hips initially.
- To progress from the A-skip, a running form emphasizes the leg recovery action with the thighs reaching parallel with the ground.



3 B-SKIP

The B-skip is a commonly prescribed sprint drill to develop the correct leg recovery and subsequent touchdown mechanics for maximum-velocity sprint running.

Starting Position

- Begin the exercise in a tall stance with the torso directly above the hips, knees, and ankles (stacked joints).

Movement Phase

- Initiate movement by lifting one leg to a flexed knee position in which the top of the thigh is near parallel (i.e., front-side mechanics). The foot of the lifted leg (swing leg) should be approximately at knee height of the stance leg, with the heel pulled up under the buttocks and the ankle dorsiflexed. This will result in the appearance of a figure four at the legs. The swing leg will always accord with this description.
- Begin the first skip by aggressively driving the swing leg down to the ground through an active foot, established through slight dorsiflexion. Lift the big toe up to establish appropriate dorsiflexion. (The main difference between the A- and B-skip is the more aggressive driving down of the swing leg to emphasize correct touchdown mechanics and increased angular velocities.)
- Complete the drive down of the swing leg through the forefoot until near-triple extension of the new stance leg occurs.
- The fore- to mid-foot of the new stance leg should land under the hips, maintaining the appearance of stacked joints.
- At the initiation of contact, the opposing leg should pop up rapidly to create a swing leg.
- The force generated from the active push down should coincide with a skipping motion that will result in a horizontal displacement of the body down the track.

Technique Tips

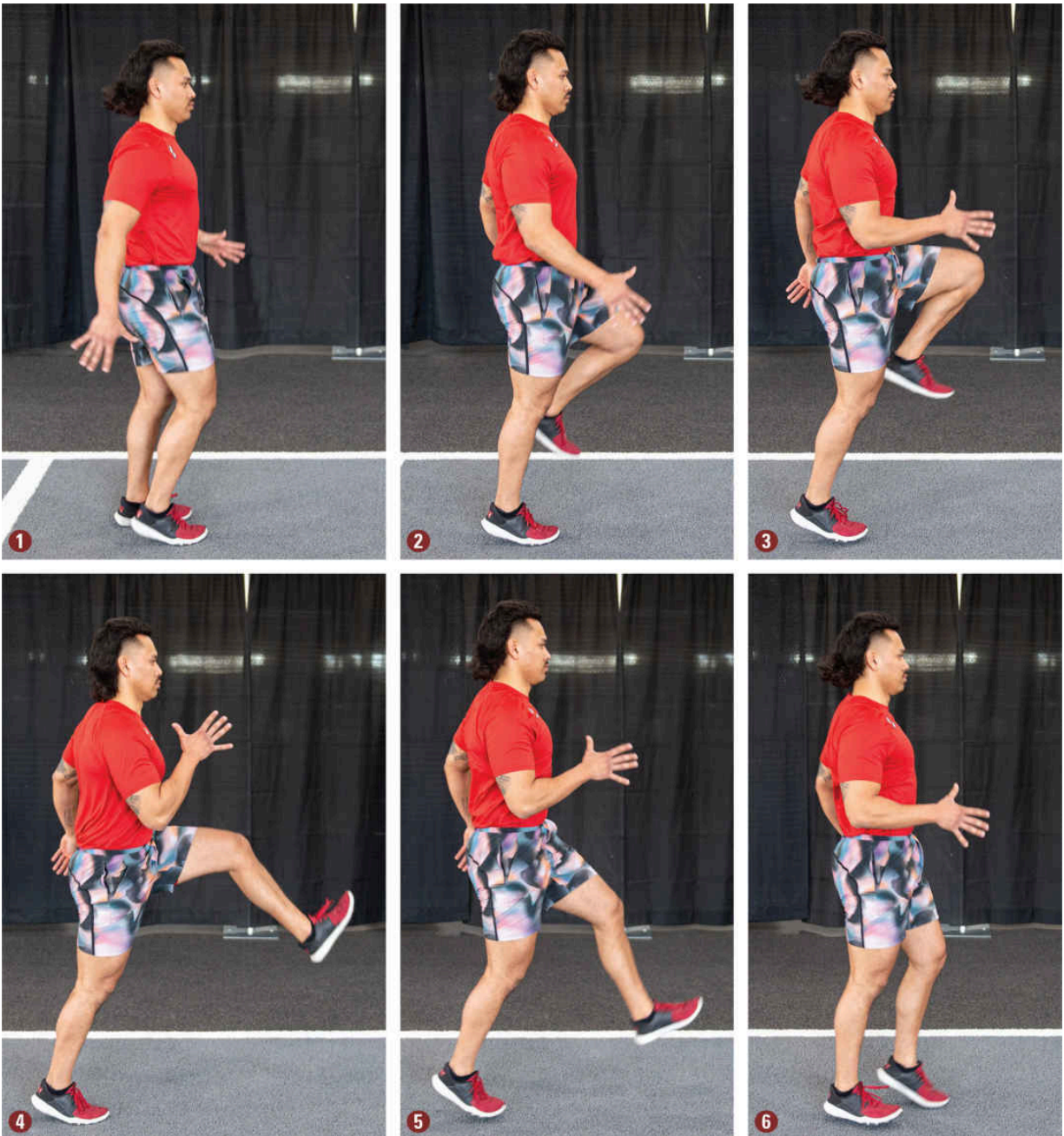
- Maintain a tall posture with relaxed shoulders through the entire duration.
- The arms should move at the same rate as the legs, with minimal to no pausing of arm action between cycles. If the athlete struggles initially to coordinate the contralateral arm

action, the athlete could place the hands on hips to initially master the correct leg recovery and touchdown action.

- During the aggressive drive down of the swing leg, ensure the leg sweeps back to contact the ground with the forefoot under the hips.

Regression and Progression

- A regression of the drill includes a B-march or -walk, which involves the same movements but at a slower pace to establish the correct leg recovery and touchdown mechanics. A progression from the B-skip involves a running form emphasizing the leg recovery to touchdown action and overhead variations to focus on trunk stability and lumbopelvic control.



4 SWITCHES

Switches emphasize rapid limb switching and path of movement of the feet that resemble the more linear path of movement associated with accelerative sprinting.

Starting Position

- Stand on one leg with one leg flexed 90 degrees at the hip and knee to ensure the thigh is parallel with the ground. The ankle is neutral or dorsiflexed, which emphasizes front-side mechanics and a positive shin angle. The contralateral arm is held with the elbow flexed to <90 degrees and with the hand around the height of the chin.
- The foot of the standing leg is flat on the ground, and the ipsilateral arm is relaxed by the side of the body.
- The trunk is upright with the head up and eyes looking forward.

Movement Phase

- While maintaining an upright trunk with the head up, the athlete simultaneously drives the standing leg up and the flexed leg down, emphasizing rapid switching of the limbs.
- The contralateral arm movement matches the speed of the leg movement.
- The landing of the previously flexed limb on the ground and the standing limb to the flexed position should occur at the same time. Pause in this position (initial position) before repeating for a desired number of repetitions.

Technique Tips

- Ensure an upright trunk and head with no leaning back or excessive lumbar flexion.
- Emphasize rapid leg and arm switching, avoiding excessively high thigh and knee lifts (>90 degrees).

Progressions

- Rather than perform with a pause between repetitions, this drill can progress to performing repeatedly for a desired number of repetitions on the spot (e.g., 2 or 3) before pausing.
- A further progression would be to add a mini-hop forward between repetitions but still emphasizing the rapid limb switching to establish a more rhythmic pattern. Trunk stability

and lumbopelvic control can be challenged with overhead variations.



5 DRIBBLES

This drill, which assists with many aspects of the technical running model involving a scaled running practice, emphasizes the more circular path of the foot during maximum-velocity sprinting.

Starting Position

- Stand with the feet hip-width apart and the head and trunk upright.

Movement Phase

- Lift one leg so the foot is above the opposing ankle with toes pointed up, then sweep the foot down and back to make foot contact. As the center of mass moves ahead of the foot, the foot should drive up and over the ankle. Repeat this circular foot motion with each limb for the designated distance. The

foot contact should be with the ankle in a neutral position and may begin heel to toe with each foot contact.

- The drill will involve a low knee lift but will become larger with progressions. The foot contact will also progress to forefoot contact with these progressions.
- The trunk leans forward slightly with the head up and eyes looking forward. The athlete adopts a contralateral arm movement at a lower amplitude of maximal sprint running, but the timing and range of motion should match the timing and range of motion of the legs.

Technique Tips

- Emphasize the circular recovery motion of the foot.
- Maintain a slight trunk lean with head up and eyes looking forward.
- Ensure the contralateral arm movement matches the timing and range of motion of the leg action.

Regressions and Progressions

- If the athlete initially struggles with the ankle dribble, he or she could lower the intensity to a walking pace and work on the circular path of movement of the foot over the opposing ankle using a heel-toe contact.
- Progression from the ankle dribble involves greater range of movement of the circular foot pattern, so the foot first reaches calf height (calf dribble) and then knee height (knee dribble).
- Furthermore, after mastering these versions, progressions could involve dribble bleeds whereby the athlete may progress from ankle to calf to knee dribble within the same repetition or blend a knee dribble to a high-speed run.



6

STRAIGHT-LEG BOUND

This drill emphasizes rapid retraction of the lower limb, offering similarity to the desired touchdown leg mechanics of maximum-velocity sprinting and increasing hip angular velocity and foot touchdown velocity.

Starting Position

- Stand with the feet hip-width apart and the head and trunk upright.

Movement Phase

- After an initial upward and forward hop from one limb, while maintaining a straight limb (extended knee and neutral ankle position), drive the opposite hip forward and then rapidly retract the limb to make ground contact on the forefoot before rebounding and repeating with the opposite limb.
- Emphasize the rapid retraction of the limb in preparation for ground contact, and ensure that ground contact times remain short and, ideally, under the hips.
- The trunk should be maintained in an upright position (must avoid leaning back), with the head up and eyes looking forward.

- The athlete should adopt a contralateral limb movement that matches the timing and range of motion of the limbs.

Technique Tips

- Maintain an upright trunk without leaning back.
- Emphasize rapid, forceful retraction of the limb toward ground contact, and ensure that ground contacts are short and under the hips.
- Adopt a contralateral arm movement that matches the timing and range of motion of the lower limbs.

Regressions and Progressions

- If the athlete initially struggles with the straight-leg bound, the athlete could lower the intensity by reducing the hip flexion range of motion and speed of retraction, and gradually and progressively build up the speed and range of motion.
- Trunk stability and lumbopelvic control can be challenged with overhead variations of straight-leg bounds, and straight-leg bounds can be transitioned or blended into normal upright running with the aim that previous straight-leg bounds will transfer to increased hip angular velocities, limb retraction, and active touchdowns with high velocities (i.e., 20-yard straight leg build up into flying 20-yard sprint).



7 SPRINT RESISTANCE: INCLINE FOR ACCELERATION

Incline sprinting is a type of resisted sprinting that is prescribed to promote physical and technical improvements within the acceleration phase of a sprint. Alternative approaches to using an incline include using weighted sleds.

Starting Position

- Begin in the correct starting position (commonly called a *crouched start*). First, split the stance position by placing the dominant leg forward and dropping the back or swing leg by one to two foot lengths.
- The length of split between the drive and swing leg is largely determined by the compromise between the (a) ability to generate sufficient forces needed to overcome a lower center of gravity and (b) comfort. In addition, the feet should be split in such a manner that the front and back legs are in line with the pelvis, so that no unnecessary twisting of the hips occurs.
- Once the split stance has been established, drop the back knee straight down so that the shank (shin) is nearer (more parallel) to the ground. This aids in promoting the proper driving position needed to begin acceleration.
- With a tall and rigid torso, raise the arm that is opposite to the front or drive leg to a position near or slightly above the forehead. The hand should be spaced approximately 6 to 8 inches (15-20 cm) away from the forehead. The rear arm (opposite to the rear or swing leg) should be pulled back to a point at which the hand is near the lateral aspect of the buttocks with an elbow angle that may range from 100 degrees to 120 degrees. This position allows for the arm action to assist in producing sufficient propulsive forces to overcome static inertia.
- On completion of setup, lean the entire body forward so that 60% of the body weight is on the front leg. Avoid breaking at the waist during this shift. The athlete should continue to feel balanced and stable.

Movement Phase

- After body weight has been transferred, the athlete can initiate the sprint through an aggressive push down and through the ground via the front or drive leg.

- The back leg should also assist in force production but will leave the ground earlier due to the starting stance position. Arm action should be synchronous with leg action.

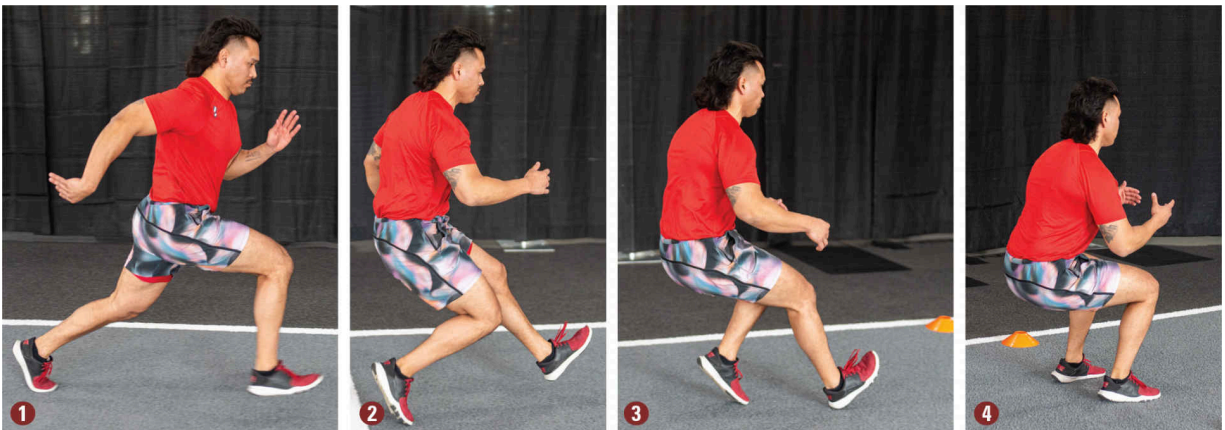
Technique Tips

- Cue the athlete to maximize accelerative mechanics through a low heel recovery of the rear or swing leg. This should be staged through the proper setup of the crouched starting position described earlier.
- Coach the athlete to continue aggressively pushing against the ground alongside an arm swing that may be cued to emphasize a hand that is pulled down and back. Arm action often parallels movements of the legs.
- The torso should remain tall and tight with the head in a neutral position. The torso and head are in line and will rise at the same rate as the hips gradually rise.

Agility Drills

8 DECELERATION DRILL

The deceleration drill requires the athlete to reduce horizontal momentum (negative acceleration) across a series of foot contacts as an isolated agility action or prior to a COD maneuver (35).



Task

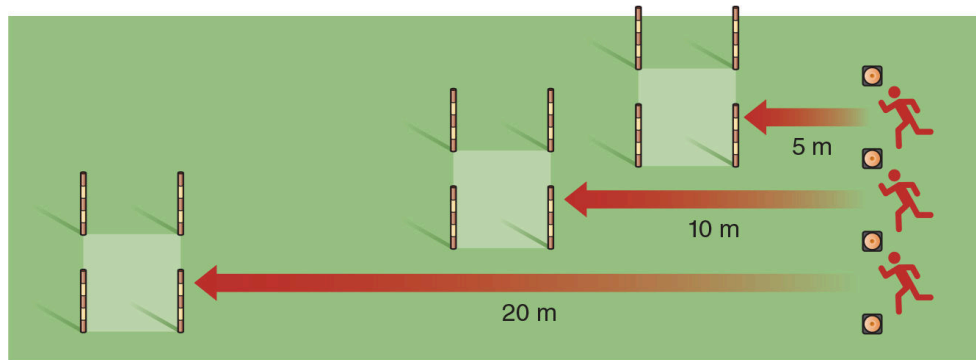
- The athlete initiates a linear acceleration from a two-point static position and sprints to a predetermined distance (i.e., 5, 10, or 20 yards or meters) before braking and decelerating over a series of steps as rapidly as possible.

Progressions and Regressions

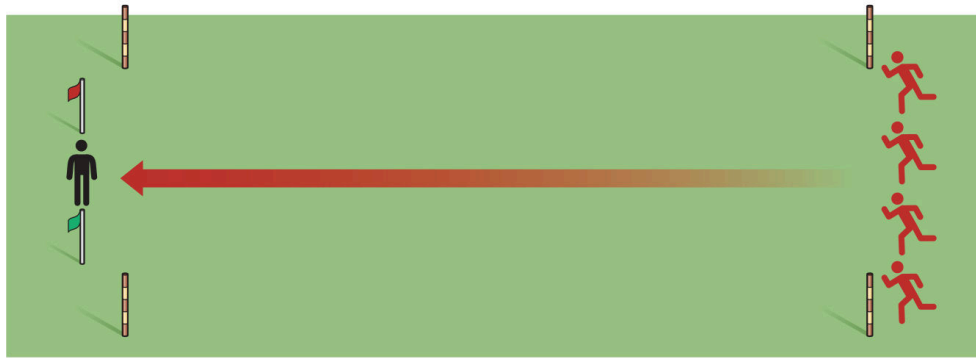
- Tasks can begin submaximally, preplanned, and over smaller approach distances and thus velocity before increasing intent and increasing approach distances and subsequent velocities, and momentum. Deceleration drills can be further developed through the use of generic and sport-specific stimuli. For example, the "Stop at the Lights" game is a game-based agility drill in which the athletes must react to the green (accelerate) and red (decelerate) signals from the coach. Athletes race to the end and are disqualified if they are unable to stop on command. Additionally, drills can be further developed through the use of auditory stimuli (i.e., "stop") and partner or opponent stimuli.

Variations

- Practitioners can vary the initiation positions and planes of motion (two-point, three-point, falling, false step, amplitude jump, crossover cut, pivot, torso rotation, drop step, pop steps) and develop a range of different deceleration actions to improve movement solutions. Deceleration positions include the following:
 - Bilateral (athletic ready position)
 - Split position
 - Transverse (jockey position)
 - Lateral deceleration position
 - Diagonal (crossover)



Setup for deceleration drill



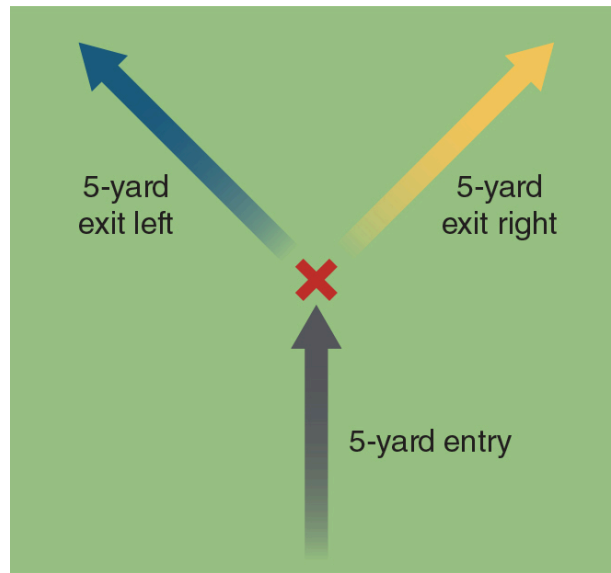
Setup for "Stop at the Lights" game

9 Y AGILITY DRILL

The Y agility drill requires the athlete to plant his or her foot laterally opposite to the direction of travel to create a propulsive impulse into the new intended direction (35). Cutting is the key movement with this drill.

Task

- The athlete initiates a linear acceleration from a two-point static position and sprints linearly to a predetermined distance (e.g., 5 yards or meters) and then performs a side-step cutting action 45 degrees to the left or right and accelerates a further 5 yards or meters (or alternative distance).



Y agility drill setup



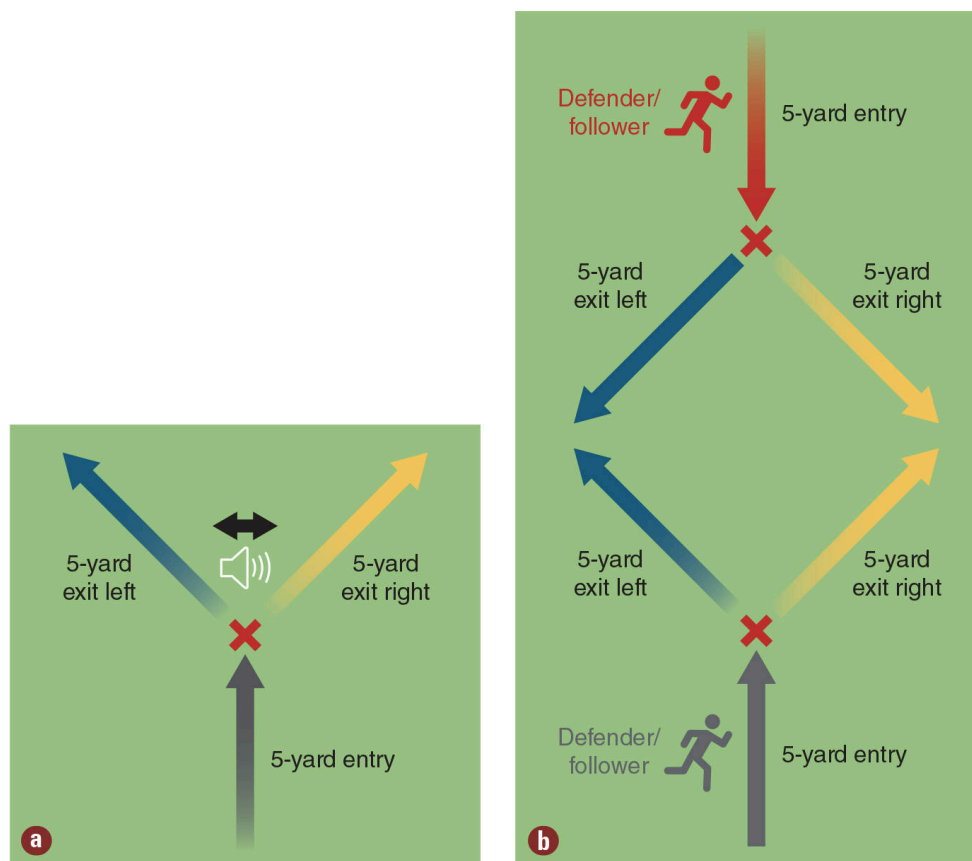
Y agility drill with sidestep cut

Progressions and Regressions

- Tasks can begin submaximal, preplanned, and applied over smaller approach distances and, therefore, involve lower approach velocities. The drill can then be progressed by performing it maximally or by increasing the approach distances, which would lead to greater approach velocities and

momentum (thereby increasing the braking demand experienced prior to and during the directional change).

- Y agility drills can be further progressed using generic or sport-specific stimuli. For example, the drill can be adapted whereby the athletes must react to auditory or visual stimuli when approaching the turning area, such as a color cone or auditory command (i.e., left or right). Moreover, the tasks can be further developed by incorporating a sport-specific stimuli such as an opponent and alternating between leading (attacking) and following (defending) through the use of mirror drills.



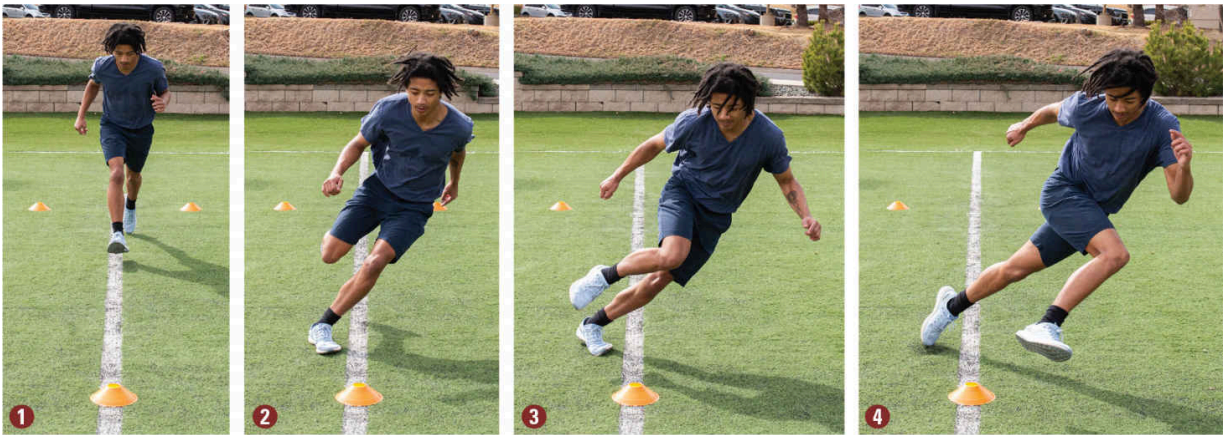
Y agility drill using (a) generic stimuli and (b) sport-specific stimuli.

Variations

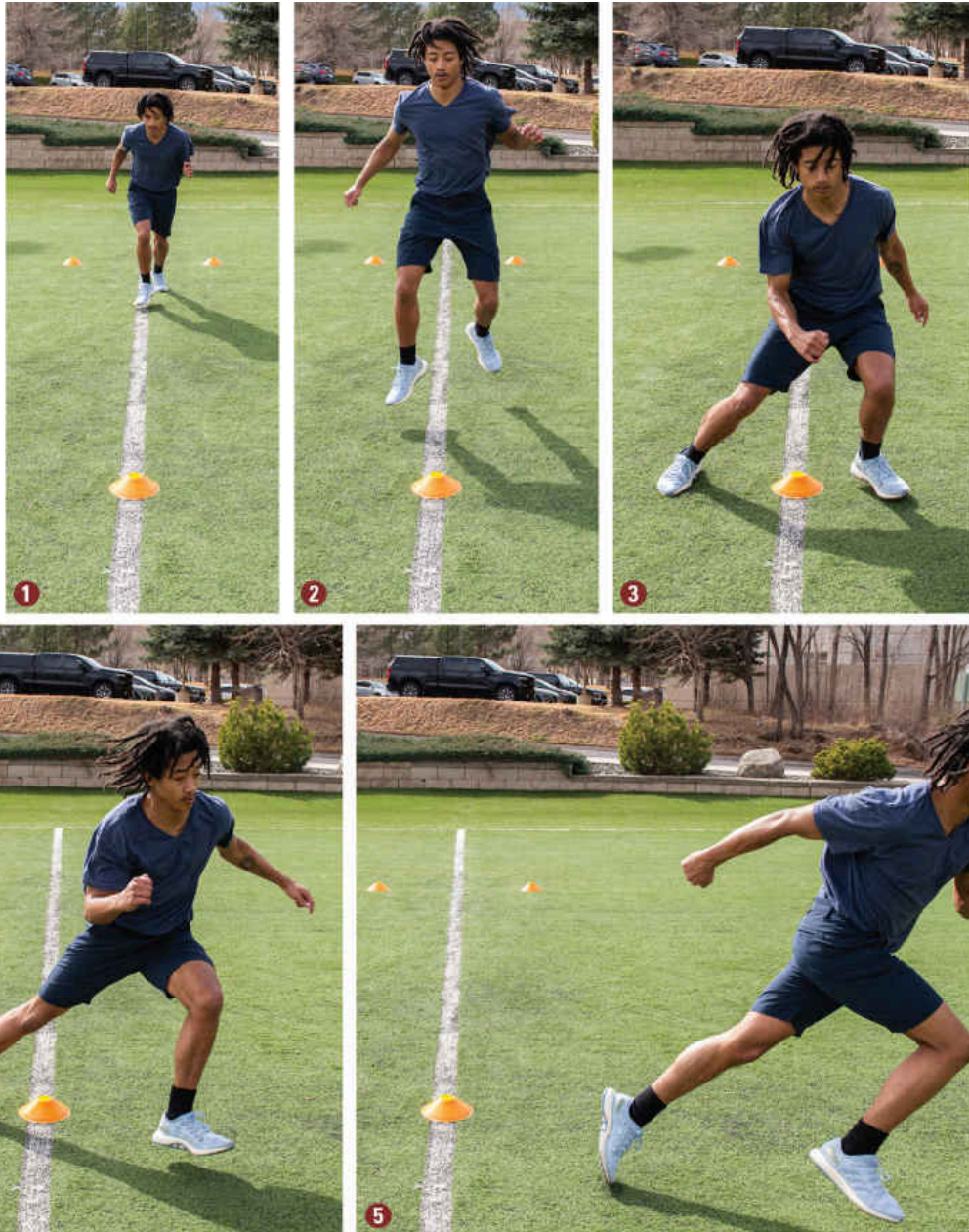
- To improve movement solutions and create between-skill variability and an adaptable performer, practitioners can vary

the cutting techniques performed at the turning point during preplanned and Y-shaped agility tasks. Alternative cuts include the following:

- *Crossover cut*. The athlete positions the plant foot on the same side (ipsilateral) of the new direction (or sometimes medially across the pelvic midline) and then crosses the opposite leg (contralateral) in front of the body for the new step in the new direction, accelerating in the same direction of the push-off leg (35).
- *Split step cut*. The athlete performs a small jump (amplitude jump) before push-off, landing with both feet greater than or equal to shoulder-width apart (not shown; it would come after the second photo), and then, on landing, the contralateral limb is used for push-off into the intended direction of travel (35).



Y agility drill cutting variation: crossover cut



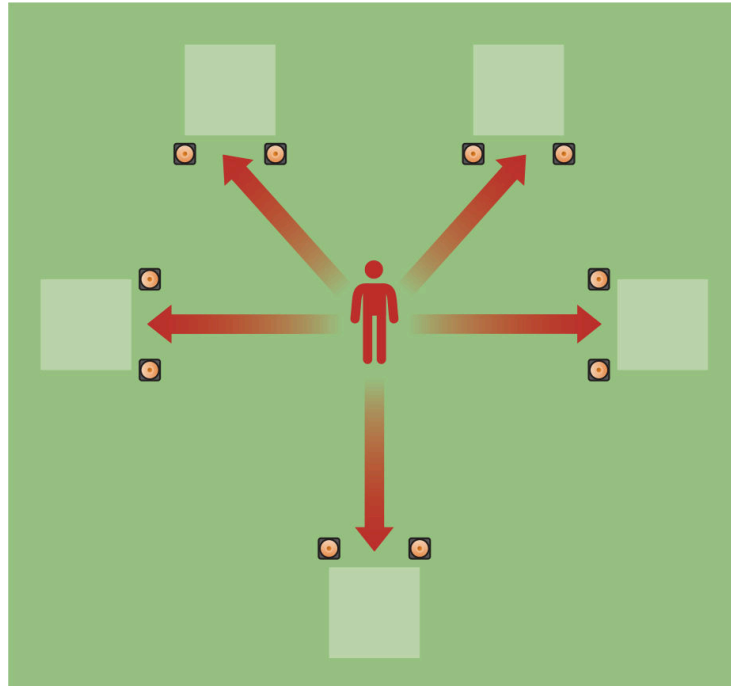
Y agility drill cutting variation: split step cut

10 MULTIPLANAR ACCELERATION INITIATION BOX DRILL

This drill requires starting movements to accelerate the athlete in a new direction (i.e., forward, sideways, rearward).

Task

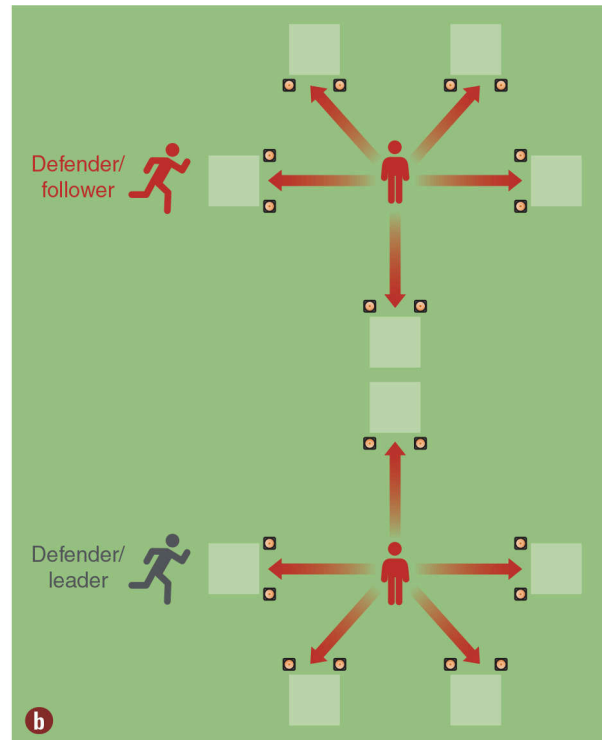
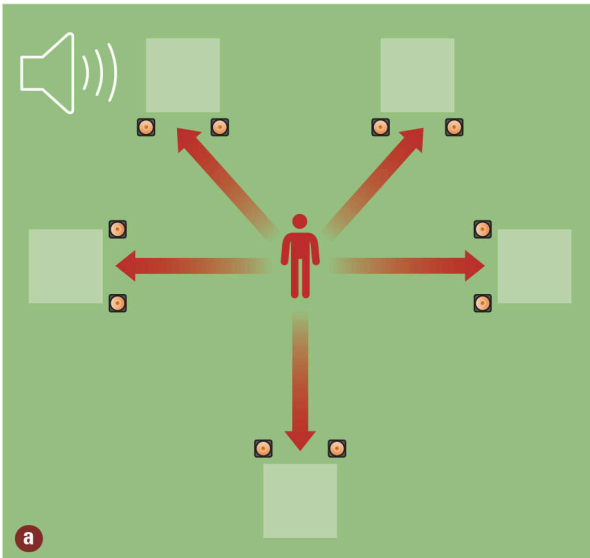
- The athlete starts in the center of a 5 × 5-yard or -meter box and initiates an acceleration from a static position in a new direction (i.e., forward, sideways, rearward) before decelerating at the specific cone or zone. The athlete should begin submaximally developing his or her ability to accelerate forward, sideways, and rearward and practice a range of different movements as outlined here:
 - *Starting and accelerating forward (static initiation)*. Bilateral to false step, two-point staggered, three-point stance, prone, kneeling, supine, two-point staggered with torso rotation
 - *Starting and accelerating sideways (static initiation)*. Jab step (side-step cut), split step (amplitude jump), shuffle step, crossover or pivot, kneeling, two-point staggered with torso rotation
 - *Starting and accelerating rearward (static initiation)*. Pivot, drop or pop turn, turns from nondominant and dominant sides



Setup for the multiplanar acceleration initiation box drill

Progressions and Regressions

- Multiplanar acceleration initiations to decelerations can be further developed using generic and sport-specific stimuli. For example, multiplanar acceleration initiation box drills can be progressed whereby the athletes must react to an auditory or visual stimulus when initiating their acceleration, such as a color cone or auditory command (e.g., left, right, forward, rear left, rear right) (figure *a*). The tasks can be further developed by incorporating a sport-specific stimuli such as an opponent and alternating between leading (attacking) and following (defending) through the use of mirror drills whereby the defender has to mirror and copy the initiation movement and direction of travel (figure *b*).



Multiplanar acceleration initiation box drill using (a) generic stimuli and (b) sport-specific stimuli.

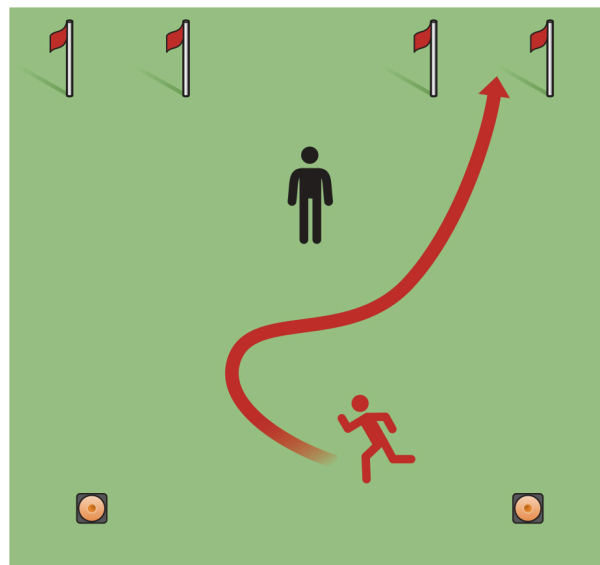
11 EVASION DRILL

Task

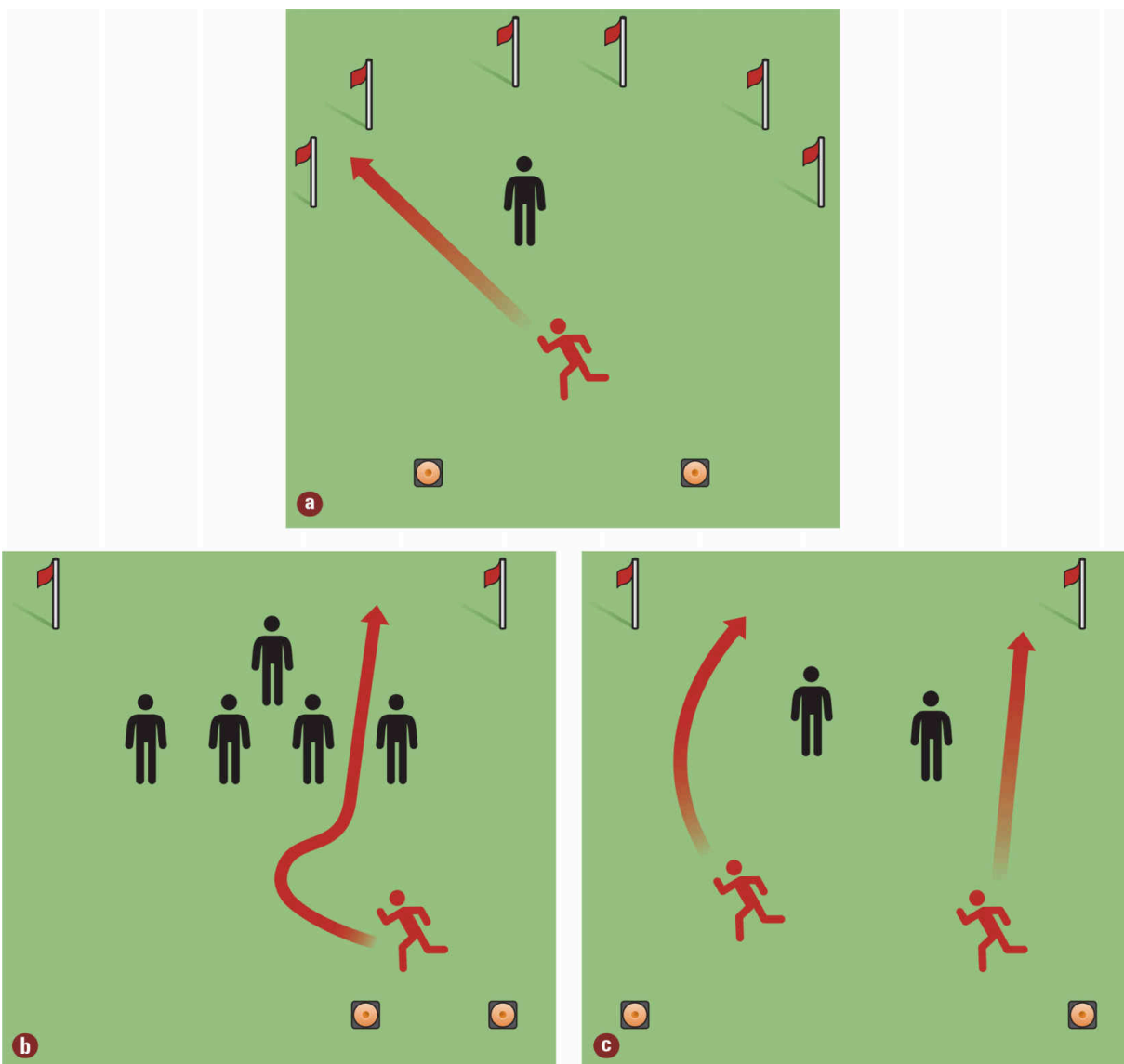
- Using a 10 × 10-yard (or -meter) box, the purpose of this open drill from an attacking perspective is to create space, take space, create separation, evade an opponent, and penetrate defensive lines accurately and efficiently and entering a specific end zone (e.g., 1v1 evasive drill with two gates). A range of different COD actions, deceleration actions, or simply linear or curvilinear speed actions could be performed to achieve the objective. Conversely, from a defensive perspective, the aim is to prevent the attacker from penetrating the defensive line and entering the end zones, limiting space for the attacker, potentially making a tag, or delaying the opponent. The defender is encouraged to identify kinematic and postural cues and anticipate and react to the opponent.

Progressions and Regressions

- A range of different mechanical and perceptual-cognitive factors can be manipulated within agility drills, which can create variation, new learning environments, and problems to solve while also influencing the task complexity and cognitive loading, which are summarized in [table 20.7 \(page 666\)](#). For example, the pitch dimensions could be expanded (figure *a*), which can increase the entry and exit velocities and momentums with a variation in end zones for the attacker to target. To increase game representativeness, the addition of static mannequins or defenders can further replicate the chaotic environments that are often encountered in sports such as American football (figure *b*). Finally, most multidirectional sports have multiple attackers and defenders; thus, to increase specificity, 2v2 small-sided games (and other configurations) also can be used to work on attacking and defensive transitions, organization, and communication (figure *c*).



Setup for the evasion drill: 1v1 (two gates)



Variations of the evasion drill: (a) 1v1 (three gates); (b) 1v1 (static defenders); (c) 2v2 (static defenders)

KEY TERMS

acceleration
 actualization
 agility
 backpedaling
 backward running
 braking impulse
 braking phase

change of direction
coordinative overload
deceleration
dynamic correspondence
force
ground contact time
ground preparation
impulse
impulse–momentum relationship
initiation
maneuverability
momentum
negative acceleration
perception–action coupling
perceptual-cognitive
periodization
propulsive phase
rate of force development (RFD)
recovery
scalar quantity
speed
spring–mass model (SMM)
sprinting
step frequency
step length
strength
stretch-shortening cycle (SSC)
stride
transition
vector quantity
velocity

STUDY QUESTIONS
